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लालची

Greedy

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Arrays & Sorting

Prerequisites - arrays

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What is Greedy Algorithm?

0/1 knapsack

↓

Greedy ✗

Angan Dakait

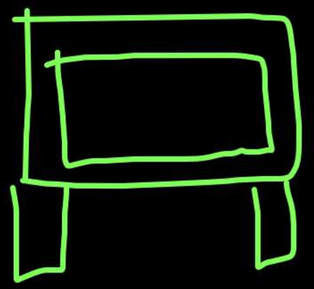
	a	b	c	d	e	f	g
	3	4	3	3	4	2	9
	22	27	22	21	26	13	54
	7.33	6.75	7.33	7	6.5	6.5	6

Bag Capacity - 10

max Loot = 71

Greedy → a , c , d → 65

3+3+3



2 Lac Dining Table

1 Lac ~~total~~ Chain Silver

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Ques: N meetings in a room

one room

1-2
2-3] α

Simple

start[] = [1, 3, 0, 5, 8, 5], end[] = [2, 4, 6, 7, 9, 9]

pair = {(1,2), (3,4), (0,6), (5,7), (8,9), (5,9)}

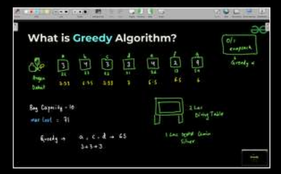
↓

sort on end time

pair = {(1,2), (3,4), (0,6), (5,7), (8,9), (5,9)}

count = 1 2 3 4

endTime = 2 4 7 9



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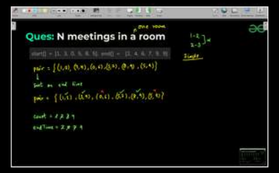
HW: Non overlapping intervals



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Ques: Assign Cookies

greed = { 5, 4, 1, 3, 4 }

↓ sort

0	1	2	3	4
1,	3,	4,	4,	5
				j

cookies = { 7, 6, 2, 1, 3, 3, 2 }

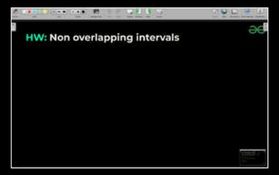
↓ sort

{ 1, 2, 2, 3, 3, 6, 7 }

i

two-pointer approach

count = j



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Ques: Boats to save people

people = { 1, 2, 3, 6 } limit = 6 boats = 3

i
j

people = { 5, 2, 3, 1 } limit = 6 boats = 2

↓ sort

{ 1, 2, 3, 5 }

j i

p = { 1, 4, 5 } limit = 6 boats = 2

i
j

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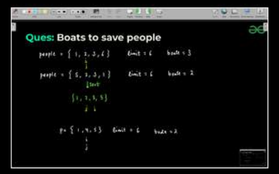
$p = \{1, 5, 6\}$ limit = 6 boats = 2

i j

```
if (p[i] + p[j] > limit) j--;
```



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Ques: Jump Game

2 3 1 0 4 true

	0	1	2	3	4	
$\{n-1\}$	2	3	1	0	4	maxPos = ~~0~~ ~~1~~ 4 true
$\{n-2\}$	2	2	1	0	4	maxPos = ~~0~~ ~~1~~ 3 false
$\{n-3\}$	1	0	8	1	2	maxPos = ~~0~~ 1 false

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Ques: Fractional Knapsack

Angan Dakait

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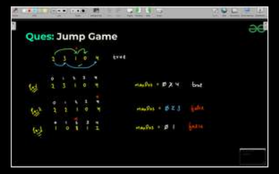
	22	27	22	21	26	13	54
	7.33	6.75	7.33	7	6.5	6.5	6

Capacity - 11

Max Loot = a + c + d + b/2

3 + 3 + 3 + 4/2 = 11

22 22 21 + 13.5 = 78.5



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BST-2

OOPS-2

THANKYOU

Cuties